A Poissonous Chicago?

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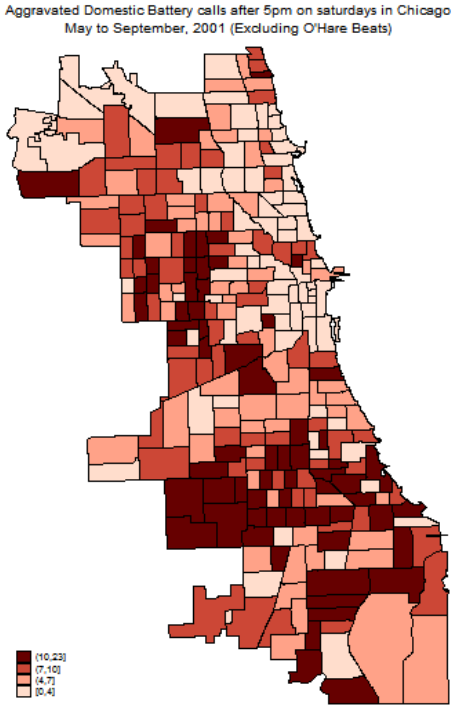
Outline of Presentation and Idea

- This presentation you are going to find that is the presentation of Prof. Koenker $-\epsilon$
- Apply Empirical Bayes machinery to understand Domestic Battery Calls in Chicago
- This is part of a larger project studying the effects *immediate emotions*, or *visceral factors:*
  - We use sports events as triggers of such type of emotions
- The idea was to explore the Empirical Bayes applied to Poisson processes, which is a fairly recent topic studied by Brown, Greenshtein and Ritov (2013)

Motivation



Descriptive Statistics: Battery Calls.

Saturday-Sundays per Month in every Chicago Beat

	N	mean	Std. Dev.	Min.	Max.
Battery	37467	8.59	5.59	0	48
Mean	37467	51.96	19.51	-7	91
Temperature					

Beats: 311 beats over 12 years (2001-2012)

The dataset on Crime is extracted from the Chicago Police Department's CLEAR (Citizen Law Enforcement Analysis and Reporting) system. It contains reported incidents of crime (with the exception of murders) that occurred in the City of Chicago from January 2001 to December 2012.

Empirical Bayes for Poisson

The usual approach is to minimize the squared error loss function, where the overall expected loss of an estimator $\delta(x)$ of λ is

$$W(\delta) = \int \sum_x (\delta(x) - \lambda)^2 p(x|\lambda)\, dG(\lambda)$$

where $p(x|\lambda) = \left(e^{-\lambda}\lambda^x\right)/x!$ and $x = 0, 1, \ldots$. The EB for λ_i given an observation $X_i = x$ is:

$$\delta_G(x) = \frac{(x+1)\, p_G(x)\, (x+1)}{p_G(x)}$$

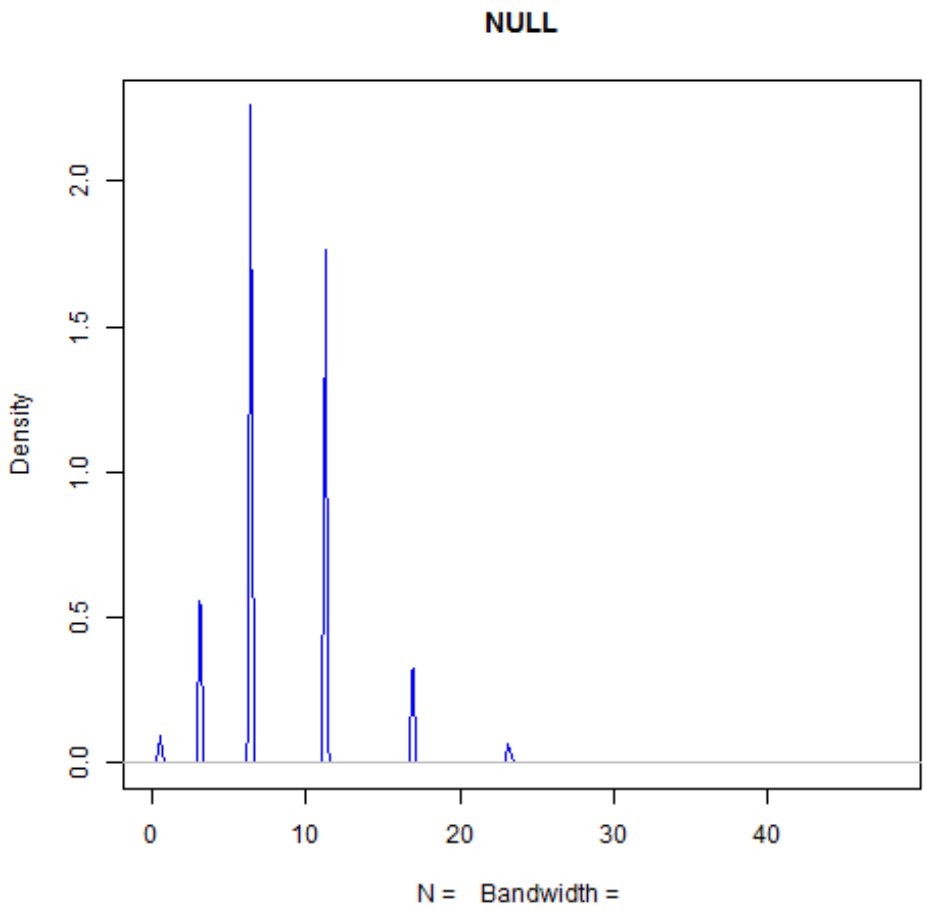
Now we may estimate $p_G(x)$ trivially by the empirical distribution $\hat{p}_G(x) = \frac{\#\{i|Y_i=y\}}{n}$

$$\hat{\delta}_G(x) = \frac{(x+1)\, \hat{p}_G(x)\, (x+1)}{\hat{p}_G(x)}$$

Empirical Bayes Results: Poisson

Kiefer and Wolfowitz estimator has the advantage that estimates directly $\hat{\delta}_G(x)$.

Results on the subsample for the years 2001-2011. (2012 reserved for forecasting)



System time: user: 0.56; system, 0.01; elapsed : 0.37

Empirical Bayes Results: Normal transformation

Brown et al (2013) suggest that we can apply the normal EB procedure to the Poisson problem easily.

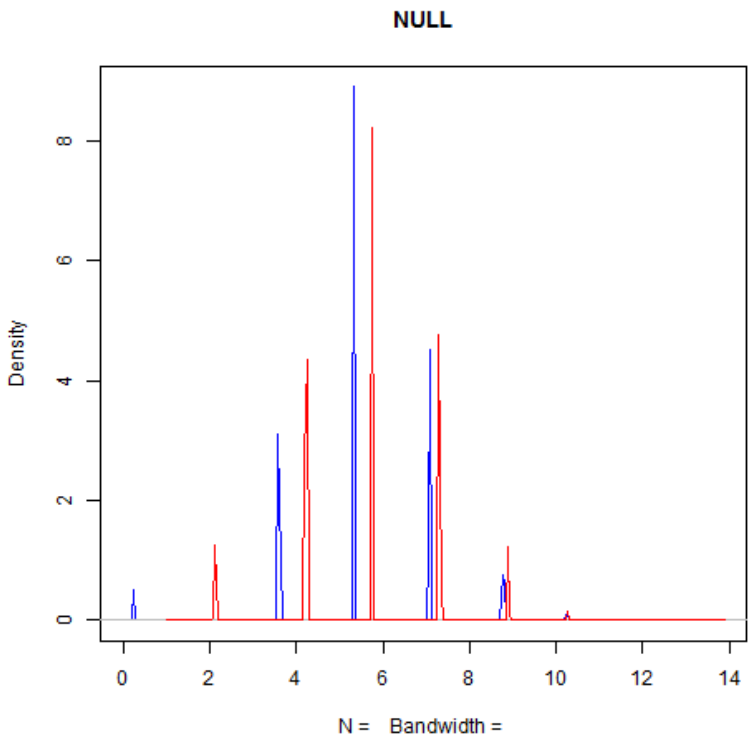
1. First transform the Poisson variables $Z_i = 2 * \sqrt{X + q}$
2. Choose and appropriate q. The most common is $q = 0$, but Brown recommends $q = 0.25$
3. Treat Z_i as $N\left(2\sqrt{\lambda}, 1\right)$, estimate the means
4. Finally get $\hat{\lambda}_i = \frac{1}{4}\hat{\mu}_i^2$

The EB to estimate $\hat{\mu}$ is Tweedie's formula

$$\hat{\delta}_N(x) = x + \sigma^2 \frac{\hat{g}'(x)}{\hat{g}(x)}$$

And we know how to do that

Empirical Bayes Results: Normal transformation




```
Call: Martin and Walker
 density.default(x = p2)

Data: p2 (37467 obs.); Bandwidth 'bw' = 0.2172

       x                 y
 Min.   :-0.3543   Min.   :9.800e-07
 1st Qu.: 3.3746   1st Qu.:2.389e-03
 Median : 7.1036   Median :3.429e-02
 Mean   : 7.1036   Mean   :6.698e-02
 3rd Qu.:10.8325   3rd Qu.:1.310e-01
 Max.   :14.5614   Max.   :2.119e-01
> p3

Call: Kiefer-Wolfowitz
 NULL

Data:  ( obs.); Bandwidth 'bw' =

       x                   y
 Min.   :-0.000001   Min.   :0.00000
 1st Qu.: 3.464101   1st Qu.:0.00000
 Median : 6.928203   Median :0.00000
 Mean   : 6.928203   Mean   :0.07193
 3rd Qu.:10.392305   3rd Qu.:0.00000
 Max.   :13.856407   Max.   :8.89579
```

blue q=0; red q=0.25

System time: K-W (GLmix) user: 342.03; system, 4.06; elapsed : 184.07.

M-W: user: 1391.42; system, 0.73; elapsed : 1417.85

To Do

- Compare how K-W performs in a real data set compared to other estimators in the literature (à la Koenker (2013)- A Gaussian compound decision bakeoff)
- How it performs in terms of Forecast using 2012 data
- Introduce covariates. (Have no idea yet how to program that)

Comments? Suggestions?